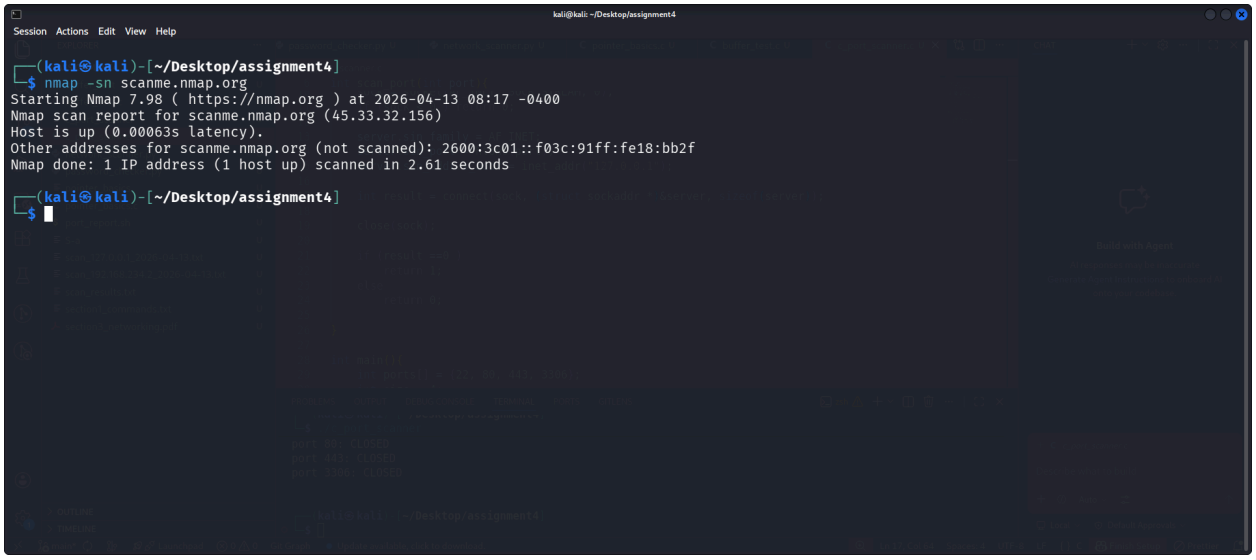
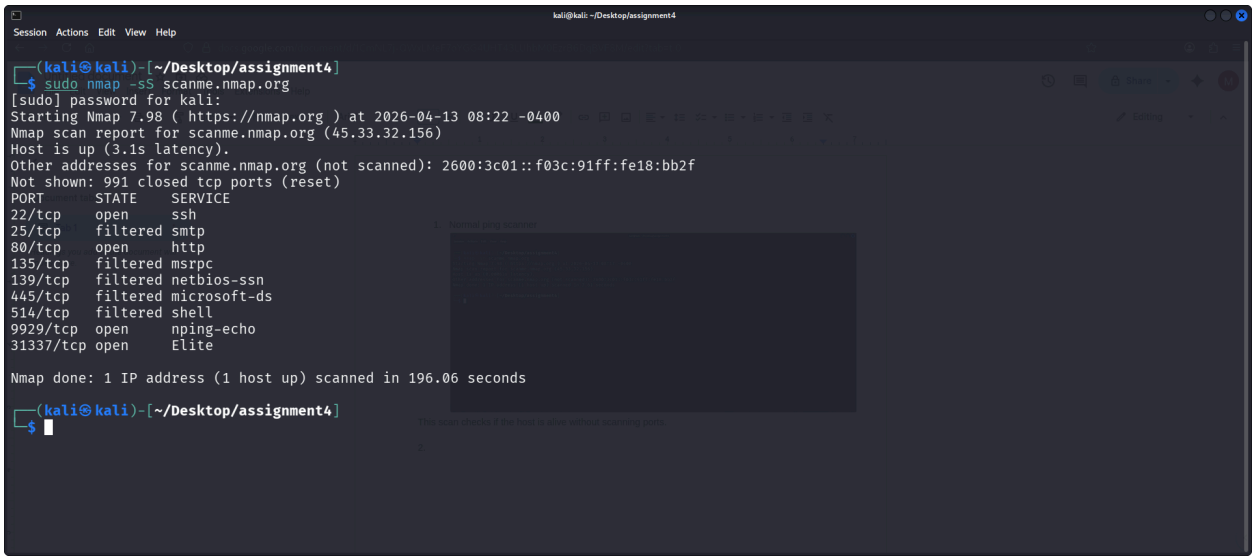
1. Normal ping scanner



nmap -sn scanme.nmap.org
This scan checks if the host is alive without scanning ports.

2.tcp syn scan



Sudo nmap -sS scanme.nmap.org
This is a stealth scan used to identify open ports on a target without completing a full connection.

3.service version detection

```
kali@kali: ~/Desktop/assignment4
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~/Desktop/assignment4]
└─$ nmap -sV -p 80 scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-13 08:29 -0400
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.00061s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT   STATE    SERVICE VERSION
80/tcp filtered http

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.16 seconds

┌──(kali㉿kali)-[~/Desktop/assignment4]
└─$
```

Nmap -sV -p 80 scanme.nmap.org
This scan identifies the service and version running on a specific port (port 80 in this case).

4. Os detection

```
kali@kali: ~/Desktop/assignment4
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~/Desktop/assignment4]
└─$ sudo nmap -O scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-13 08:32 -0400
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
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RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.59s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 991 closed tcp ports (reset)
PORT      STATE    SERVICE
22/tcp    open     ssh
25/tcp    filtered smtp
80/tcp    open     http
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
514/tcp   filtered shell
9929/tcp  open     nping-echo
31337/tcp open     Elite
Aggressive OS guesses: Actiontec MI424WR-GEN3I WAP (98%), DD-WRT v24-sp2 (Linux 2.4.37) (98%), Microsoft Windows XP SP3 or Windows 7 or Windows S
erver 2012 (96%), Linux 3.2 (95%), Linux 4.4 (95%), Microsoft Windows XP SP3 (94%), BlueArc Titan 2100 NAS device (90%), VMware Player virtual NA
T device (89%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 227.74 seconds

┌──(kali㉿kali)-[~/Desktop/assignment4]
└─$
```

Sudo nmap -O scanme.nmap.org
This scan attempts to detect the operating system of the target using network fingerprinting techniques.